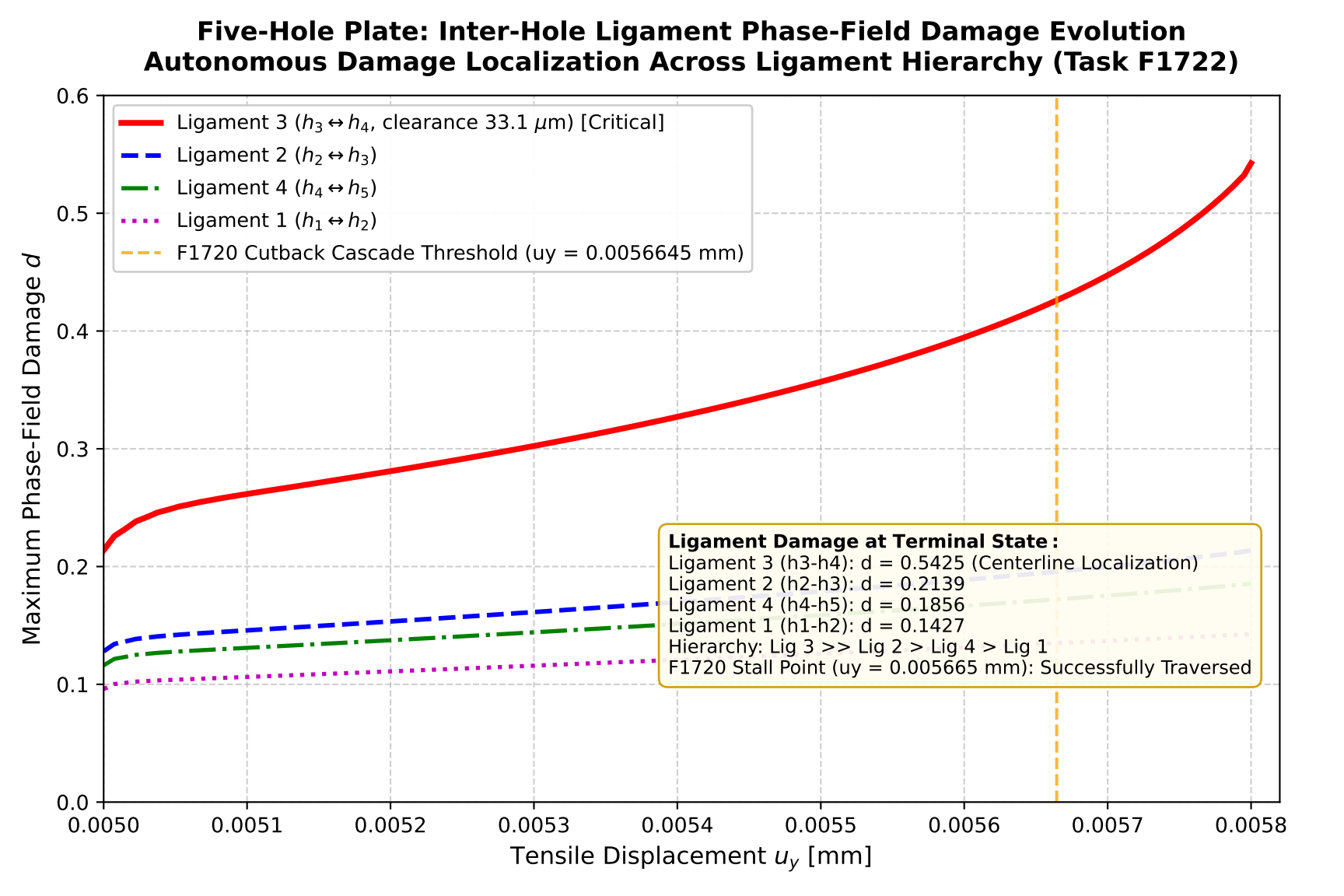
Five-Hole Plate: Inter-Hole Ligament Phase-Field Damage Evolution
Autonomous Damage Localization Across Ligament Hierarchy (Task F1722)
Ligament 3 ($h_3 \leftrightarrow h_4$, clearance 33.1 μm) [Critical]
Ligament 2 ($h_2 \leftrightarrow h_3$)
Ligament 4 ($h_4 \leftrightarrow h_5$)
Ligament 1 ($h_1 \leftrightarrow h_2$)
F1720 Cutback Cascade Threshold (uy = 0.0056645 mm)
Ligament Damage at Terminal State :
Ligament 3 (h3-h4): d = 0.5425 (Centerline Localization)
Ligament 2 (h2-h3): d = 0.2139
Ligament 4 (h4-h5): d = 0.1856
Ligament 1 (h1-h2): d = 0.1427
Hierarchy: Lig 3 >> Lig 2 > Lig 4 > Lig 1
F1720 Stall Point (uy = 0.005665 mm): Successfully Traversed
Maximum Phase-Field Damage d
Tensile Displacement u_y [mm]
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.0050
0.0051
0.0052
0.0053
0.0054
0.0055
0.0056
0.0057
0.0058